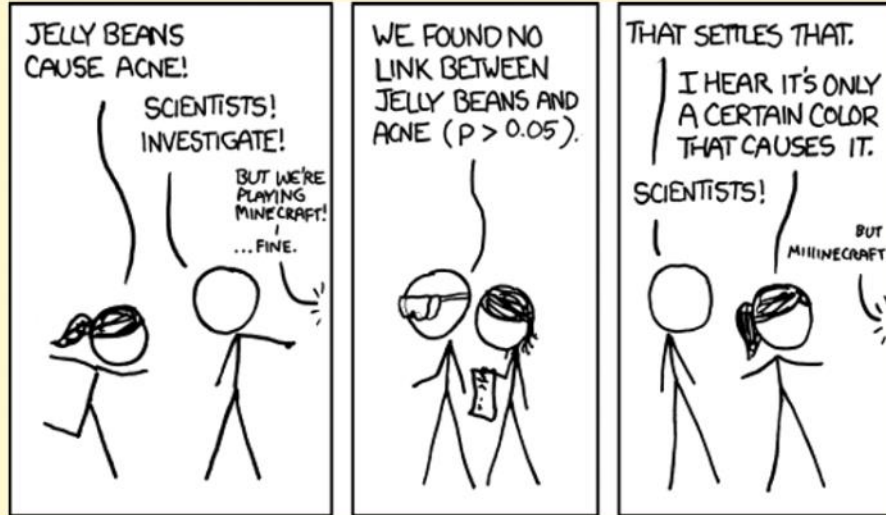


CV 5101  
**Modeling, Uncertainty, and**  
**Data** for Engineers  
(July – Nov 2026)

Dr. Prakash S Badal

L-13 (HT)  
Sep 25, 2026

xkcd

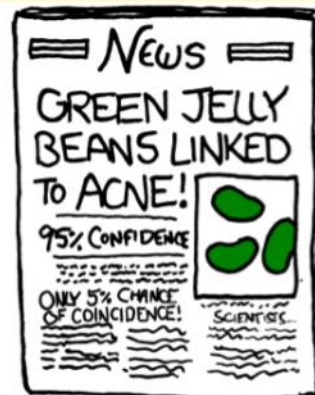


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<https://explainxkcd.com/882>

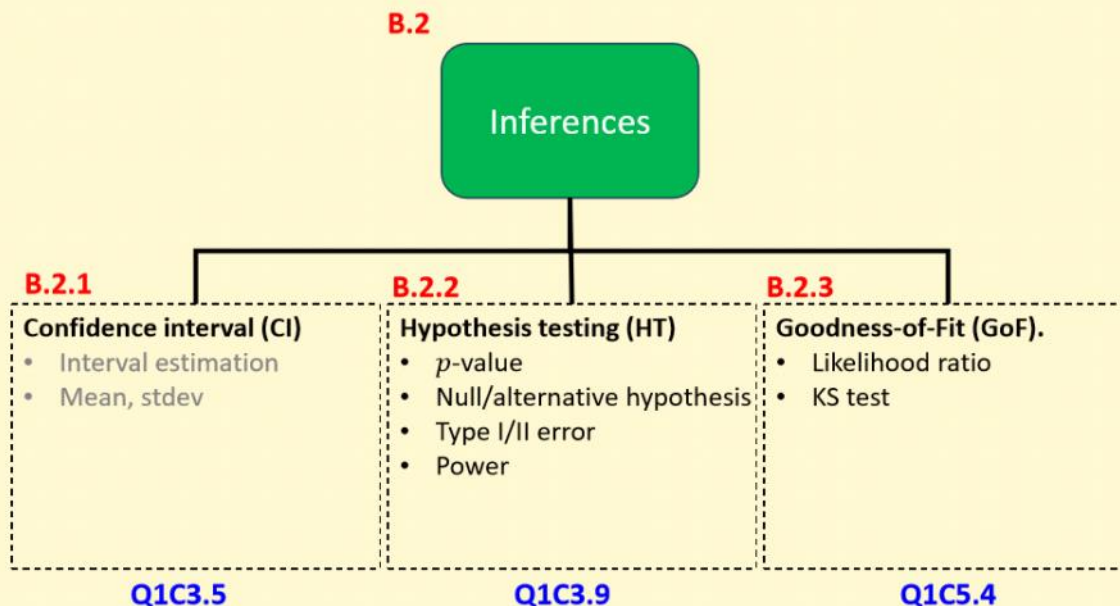


# Parameter estimation

- Next remaining classes:
  - Estimation: MoM, MLE
  - Inference: Confidence interval (CI)
  - **Inference**: Hypothesis testing,  $p$ -value, power
  - **Goodness of fit (GoF)**:  $\chi^2$ , KS
- Risk & reliability (Q2C8)
  - Introduction
  - Risk Evaluation
- **Extreme value analysis (Q2C7)**
  - Peak over threshold
  - Return period and design life

5

## Module Overview



# Flow

- Null/alternative hypothesis
- Type I/II error
- Hypothesis testing method
- Error calculations
- Power of test
- $p$ -value
- GoF

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## Central limit theorem: recap

- For large sample size  $n$ , **sample mean** approaches

a **normal distribution**,

with **mean  $\mu$**  (same as population mean), and

**variance  $\sigma^2/n$**  (less than population Var.).

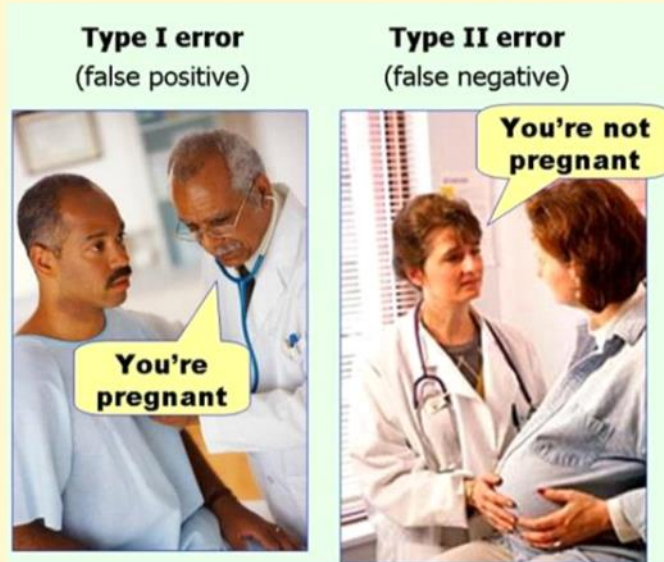
**Spreadsheet demo!**

8

# Spot the error\*

\*: not an example from real life

false positive ↗  
 Accuracy of the test ↗  
 false negative ↘  
 Result of the test ↘



People are NOT generally pregnant.  
 NULL HYPOTHESIS  
 $H_0$ : YOU ARE NOT PREGNANT → default  
 $H_1$  (ALTERNATIVE): YOU ARE PREGNANT.

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## Justice system in a democracy

- How can someone be arrested if they really are **presumed innocent**?
- Why is a defendant pronounced "**Not Guilty**" instead of innocent?
- Why do citizens put up with a system that allows **criminals to go free on technicalities**?

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## Common challenges in justice, research (life?)

→ **Strong Urge** to believe in an unusual events:

e.g. \* A new medicine cures cancer.

\* A charge person (aka 'defendant') is guilty

→ **Serious Repurcussions** of being wrong

e.g. \* Two types of wrong decisions are possible

| YOU ACTED ON YOUR URGE, BUT IT TURNED OUT TO BE FALSE | YOU DID NOT ACT ON YOUR URGE, AND IT TURNED OUT TO BE TRUE  |
|-------------------------------------------------------|-------------------------------------------------------------|
| → innocent person goes to jail                        | → guilty persons goes free                                  |
| → Patient get wrong medicine                          | → Patients dying/suffering, despite a cure being available. |
| → No Sure shot way to prove the unusual event.        |                                                             |
| → Limited Data                                        |                                                             |

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# Hypothesis testing is the solution!

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# Hypothesis testing (& justice system)

No numerical values in courts, but they share four common features:

1. **NULL HYPOTHESIS:**  $H_0$   
The presumption of innocence.
2. **ALTERNATE HYPOTHESIS:**  $H_1$   
Why a suspected person is arrested.
3. **STANDARD OF JUSTICE:** significance level,  $\alpha$   
A reasonable doubt.
4. **A DATA SAMPLE:** Test score.  
Evaluation of partial information.

# Hypothesis testing (& justice system)

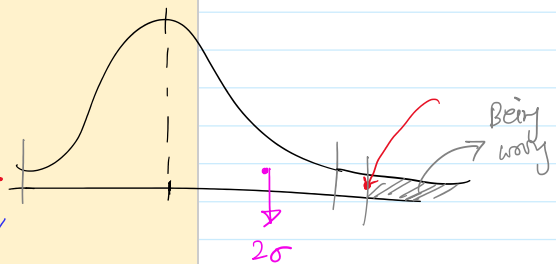
Both statistical testing and the justice system:

1. Concentrate on **rejecting the null hypothesis**

- It is easier
- Rejecting "Presumption of Innocent".  
 $\equiv$  "defendant is pronounced guilty"

2. Consider a failure to reject **the null hypothesis**

- As "Not Guilty" verdict
- Proving  $H_0$  is true will take endless evidence.



# Hypothesis testing (& justice system)

Neither statistical testing and the justice system are perfect:

- Sometimes, the jury makes an error.
  - An **innocent person goes to jail**
  - Statisticians call it a **Type I**.
- Sometimes, **a guilty person is set free**
  - Statisticians call it a **Type II**.

Which ONE is worse.

Citizens find Type II disturbing but is NOT as horrifying as Type I error.

Type I error = { An Innocent person goes to Jail  
+  
A guilty person walks free.

L-2/2 (HT)  
Sep 30, 2026

# Hypothesis testing (& justice system)

No numerical values in courts, but they share four common features:

## 1. The alternative hypothesis:

This is why a *criminal is arrested* (or a research undertaken)  
denoted by  $H_1$  or  $H_A$ .

## 2. The null hypothesis:

The *presumption of innocence* (the solution won't work!)  
 $H_0$  is the logical opposite of  $H_1$ .

## 3. A standard of justice:

Absolute proof IMPOSSIBLE, hence a *reasonable doubt* (sig. level)  
Reject the null hypothesis beyond a reasonable doubt.

## 4. A data sample:

Evaluation of *partial information* (test score)  
The "whole truth and nothing but the truth" is often impossible.

# Errors in Hypothesis testing (& justice system)

$$\text{Type I error} \equiv \left\{ \begin{array}{c} \text{an innocent person goes to jail} \\ + \\ \text{a guilty person walks free} \end{array} \right\}$$

HT (& justice system) puts a lot of emphasis on  
avoiding Type I error.

Counter example: Airport security

Innocent person and alarm goes off (Type I error)

Guilty person and alarm goes not go off (Type II error)

**Sometimes, Type II error is worse!**

e.g., airport security.

## Errors in Hypothesis testing (& justice system)

- An innocent person goes to jail *Type I*
- A guilty person is set free *Type II*

## Errors in Hypothesis testing (& justice system)

HT & justice lady put a lot of emphasis on  
**avoiding Type I error.**

### Product example:

- Null hypothesis,  $H_0$ : a product satisfies customer requirements.
- If  $H_0$  is rejected, do not sell the product to customers.

*$H_0$ : New iPhone is substantially improved.*

**Type I error:** Reject an improved model by mistake.  
*A very expensive error.*

**Type II error:** Failing to reject a bad model & shipping it to customers.  
*Loss of reputation*

**Which one is worse?** *Type II error.*

# Spot the error type

According to a report from the United States Environmental Protection Agency, burning one gallon of gasoline typically emits about 8.9 kg of CO<sub>2</sub>. A fuel company wants to test a new type of gasoline designed to have lower CO<sub>2</sub> emissions. Here are their hypotheses:

$$H_0 : \mu = 8.9 \text{ kg}$$

$$H_a : \mu < 8.9 \text{ kg}$$

(where  $\mu$  is the mean amount of CO<sub>2</sub> emitted by burning one gallon of this new gasoline).

Under which of the following conditions would the company commit a Type I error?

Choose 1 answer:

- ☐ A The mean amount of CO<sub>2</sub> emitted by the new fuel is actually 8.9 kg, and they fail to conclude it is lower than 8.9 kg.
- ☐ B The mean amount of CO<sub>2</sub> emitted by the new fuel is actually lower than 8.9 kg, and they conclude it is lower than 8.9 kg.
- ☐ C The mean amount of CO<sub>2</sub> emitted by the new fuel is actually 8.9 kg, and they conclude it is lower than 8.9 kg.
- ☐ D The mean amount of CO<sub>2</sub> emitted by the new fuel is actually lower than 8.9 kg, and they fail to conclude it is lower than 8.9 kg.

A large university is curious if they should build another cafeteria. They plan to survey a sample of their students to see if there is strong evidence that the proportion interested in a meal plan is higher than 40%, in which case they will consider building a new cafeteria.

Let  $p$  represent the proportion of students interested in a meal plan. Here are the hypotheses they'll use:

$$H_0 : p \leq 0.40$$

$$H_a : p > 0.40$$

What would be the consequence of a Type II error in this context?

Choose 1 answer:

- ☐ A They don't consider building a new cafeteria when they should.
- ☐ B They don't consider building a new cafeteria when they shouldn't.
- ☐ C They consider building a new cafeteria when they shouldn't.
- ☐ D They consider building a new cafeteria when they should.

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# Spot the error type

A quality control expert wants to test the null hypothesis that a new solar panel is no more effective than the older model.

What would be the consequence of a Type I error in this context?

Choose 1 answer:

- ☐ A They don't conclude the new panel is more effective when it actually is not more effective.
- ☐ B They don't conclude the new panel is more effective when it actually is more effective.
- ☐ C They conclude the new panel is more effective when it actually is more effective.
- ☐ D They conclude the new panel is more effective when it actually is not more effective.

According to a report from the United States Environmental Protection Agency, burning one gallon of gasoline typically emits about 8.9 kg of CO<sub>2</sub>. A fuel company wants to test a new type of gasoline designed to have lower CO<sub>2</sub> emissions. Here are their hypotheses:

$$H_0 : \mu = 8.9 \text{ kg}$$

$$H_a : \mu < 8.9 \text{ kg}$$

(where  $\mu$  is the mean amount of CO<sub>2</sub> emitted by burning one gallon of this new gasoline).

Which of the following would be a Type II error in this setting?

Choose 1 answer:

- ☐ A The mean amount of CO<sub>2</sub> emitted by the new fuel is actually 8.9 kg, and they conclude it is lower than 8.9 kg.
- ☐ B The mean amount of CO<sub>2</sub> emitted by the new fuel is actually lower than 8.9 kg, and they fail to conclude it is lower than 8.9 kg.
- ☐ C The mean amount of CO<sub>2</sub> emitted by the new fuel is actually 8.9 kg, and they fail to conclude it is lower than 8.9 kg.
- ☐ D The mean amount of CO<sub>2</sub> emitted by the new fuel is actually lower than 8.9 kg, and they conclude it is lower than 8.9 kg.

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# Example 1

- Consider bars made of IS 1786-compliant **Fe500 grade steel**
  - Let's focus on the mean yield strength
  - We are interested in deciding **whether or not** the mean yield strength is 500 MPa

Formally,

$$H_0: f_y = 500 \text{ MPa}$$

$$H_1: f_y \neq 500 \text{ MPa}$$

Four situations may arise

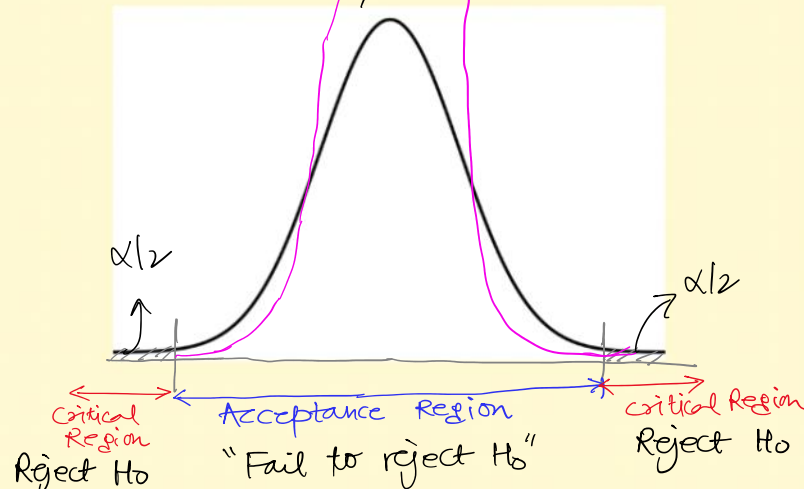
| Decision ↓           | $H_0$ is True             | $H_0$ is False            |
|----------------------|---------------------------|---------------------------|
| Fail to reject $H_0$ | No error                  | Type II error ( $\beta$ ) |
| Reject $H_0$         | Type I error ( $\alpha$ ) | No error                  |

# Example 1

significance level:  $\alpha$

$$H_0: \mu_{F_y} = 500 \text{ MPa}$$

$$H_1: \mu_{F_y} \neq 500 \text{ MPa}$$



# Type I error

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## Probability of Type I Error, $\alpha$

- $\alpha = \text{Pr}(\text{Type I error})$   
=  $\text{Pr}(\text{reject } H_0 \text{ when } H_0 \text{ is True})$

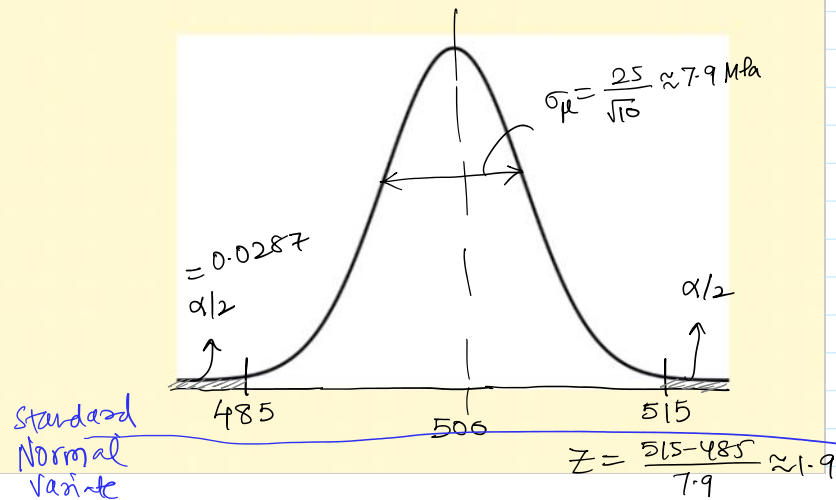
We can reduce Type I error

- by widening the acceptance region
- by increasing the number of samples

# Probability of Type I Error, $\alpha$

- Consider Fe500 bar again; suppose  $\sigma_{Fe500} = 25 \text{ MPa}$
- Suppose we tested a sample with  $n = 10$  specimens
- If a client accepts bars with average  $f_y \in [485 \text{ MPa}, 515 \text{ MPa}]$
- Find the **probability of Type I error?**

$$\alpha = 5.74\%$$



## Standard Normal Probabilities

### Z-table

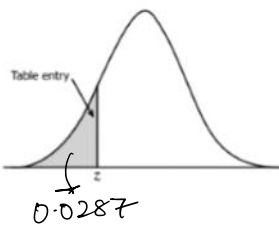


Table entry for  $z$  is the area under the standard normal curve to the left of  $z$ .

| $z$  | .00   | .01   | .02   | .03   | .04   | .05   | .06   | .07   | .08   | .09   |
|------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| -3.4 | .0003 | .0003 | .0003 | .0003 | .0003 | .0003 | .0003 | .0003 | .0003 | .0002 |
| -3.3 | .0005 | .0005 | .0005 | .0004 | .0004 | .0004 | .0004 | .0004 | .0004 | .0003 |
| -3.2 | .0007 | .0007 | .0006 | .0006 | .0006 | .0006 | .0006 | .0005 | .0005 | .0005 |
| -3.1 | .0010 | .0009 | .0009 | .0009 | .0008 | .0008 | .0008 | .0008 | .0007 | .0007 |
| -3.0 | .0013 | .0013 | .0013 | .0012 | .0012 | .0011 | .0011 | .0011 | .0010 | .0010 |
| -2.9 | .0019 | .0018 | .0018 | .0017 | .0016 | .0016 | .0015 | .0015 | .0014 | .0014 |
| -2.8 | .0026 | .0025 | .0024 | .0023 | .0023 | .0022 | .0021 | .0021 | .0020 | .0019 |
| -2.7 | .0035 | .0034 | .0033 | .0032 | .0031 | .0030 | .0029 | .0028 | .0027 | .0026 |
| -2.6 | .0047 | .0045 | .0044 | .0043 | .0041 | .0040 | .0039 | .0038 | .0037 | .0036 |
| -2.5 | .0062 | .0060 | .0059 | .0057 | .0055 | .0054 | .0052 | .0051 | .0049 | .0048 |
| -2.4 | .0082 | .0080 | .0078 | .0075 | .0073 | .0071 | .0069 | .0068 | .0066 | .0064 |
| -2.3 | .0107 | .0104 | .0102 | .0099 | .0096 | .0094 | .0091 | .0089 | .0087 | .0084 |
| -2.2 | .0139 | .0136 | .0132 | .0129 | .0125 | .0122 | .0119 | .0116 | .0113 | .0110 |
| -2.1 | .0179 | .0174 | .0170 | .0166 | .0162 | .0158 | .0154 | .0150 | .0146 | .0143 |
| -2.0 | .0228 | .0222 | .0217 | .0212 | .0207 | .0202 | .0197 | .0192 | .0188 | .0183 |
| -1.9 | .0287 | .0281 | .0274 | .0268 | .0262 | .0256 | .0250 | .0244 | .0239 | .0233 |
| -1.8 | .0359 | .0351 | .0344 | .0336 | .0329 | .0322 | .0314 | .0307 | .0301 | .0294 |
| -1.7 | .0446 | .0436 | .0427 | .0418 | .0409 | .0401 | .0392 | .0384 | .0375 | .0367 |
| -1.6 | .0548 | .0537 | .0526 | .0516 | .0505 | .0495 | .0485 | .0475 | .0465 | .0455 |
| -1.5 | .0668 | .0655 | .0643 | .0630 | .0618 | .0606 | .0594 | .0582 | .0571 | .0559 |
| -1.4 | .0808 | .0793 | .0778 | .0764 | .0749 | .0735 | .0721 | .0708 | .0694 | .0681 |
| -1.3 | .0968 | .0951 | .0934 | .0918 | .0901 | .0885 | .0869 | .0853 | .0838 | .0823 |
| -1.2 | .1151 | .1131 | .1112 | .1093 | .1075 | .1056 | .1038 | .1020 | .1003 | .0985 |
| -1.1 | .1357 | .1335 | .1314 | .1292 | .1271 | .1251 | .1230 | .1210 | .1190 | .1170 |
| -1.0 | .1587 | .1562 | .1539 | .1515 | .1492 | .1469 | .1446 | .1423 | .1401 | .1379 |
| -0.9 | .1841 | .1814 | .1788 | .1762 | .1736 | .1711 | .1685 | .1660 | .1635 | .1611 |
| -0.8 | .2119 | .2090 | .2061 | .2033 | .2005 | .1977 | .1949 | .1922 | .1894 | .1867 |
| -0.7 | .2420 | .2389 | .2358 | .2327 | .2296 | .2266 | .2236 | .2206 | .2177 | .2148 |
| -0.6 | .2743 | .2709 | .2676 | .2643 | .2611 | .2578 | .2546 | .2514 | .2483 | .2451 |
| -0.5 | .3085 | .3050 | .3015 | .2981 | .2946 | .2912 | .2877 | .2843 | .2810 | .2776 |
| -0.4 | .3446 | .3409 | .3372 | .3336 | .3300 | .3264 | .3228 | .3192 | .3156 | .3121 |
| -0.3 | .3821 | .3783 | .3745 | .3707 | .3669 | .3632 | .3594 | .3557 | .3520 | .3483 |
| -0.2 | .4207 | .4168 | .4129 | .4090 | .4052 | .4013 | .3974 | .3936 | .3897 | .3859 |
| -0.1 | .4602 | .4562 | .4522 | .4483 | .4443 | .4404 | .4364 | .4325 | .4286 | .4247 |
| -0.0 | .5000 | .4960 | .4920 | .4880 | .4840 | .4801 | .4761 | .4721 | .4681 | .4641 |

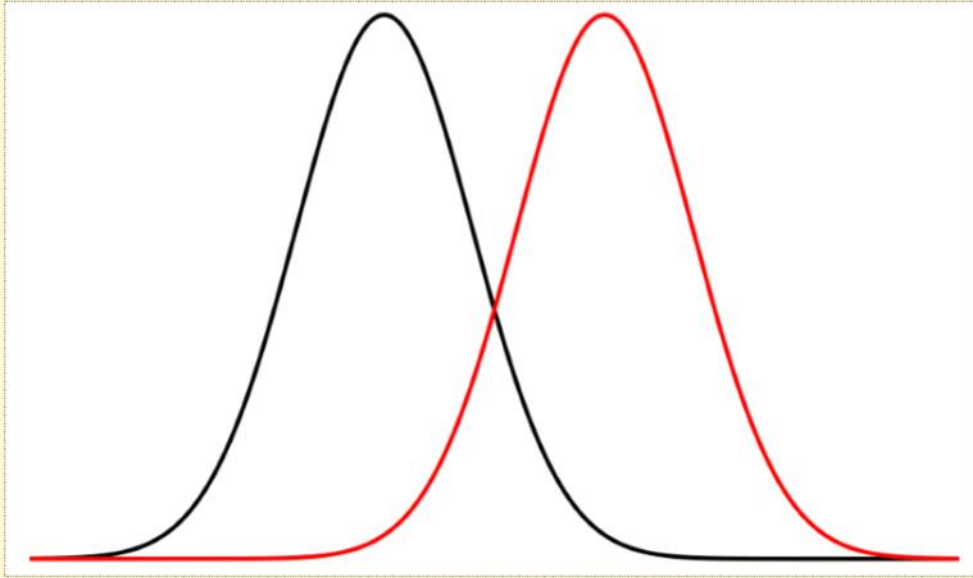
# Type II error

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## Probability of Type II Error, $\beta$

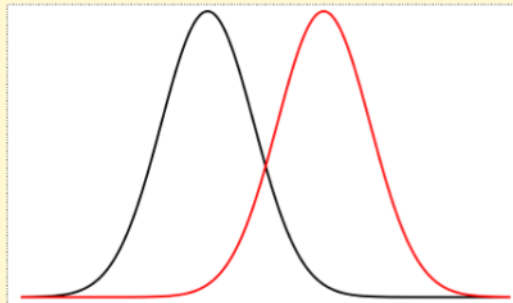
- $\beta = \text{Pr}(\text{Type II error})$   
= Pr(failure to reject  $H_0$  when  $H_0$  is False)

## Type II error



## Probability of Type II Error, $\beta$

- Fe500;  $\sigma_{\text{Fe500}} = 25 \text{ Mpa}$ ; we tested a sample with  $n = 10$  specimens
- If an alternative hypothesis exists that  $\mu_{F_y} = 520 \text{ MPa}$
- Find the **probability of Type II error**?



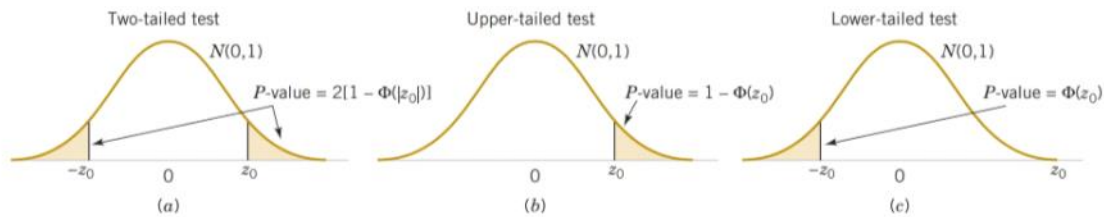
# Power of a statistical test

- The **power** of a statistical test

## Power: steel strength example

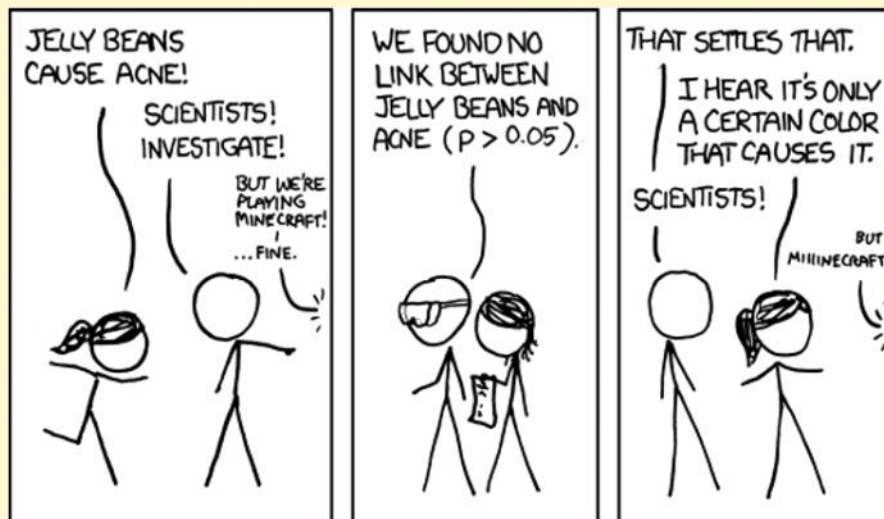
- In the steel strength problem, we found that  $\beta = 0.2643$  when the true mean is  $\mu = 520$ , what is the power of this test

# p-value



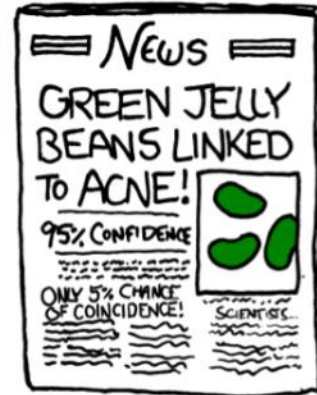
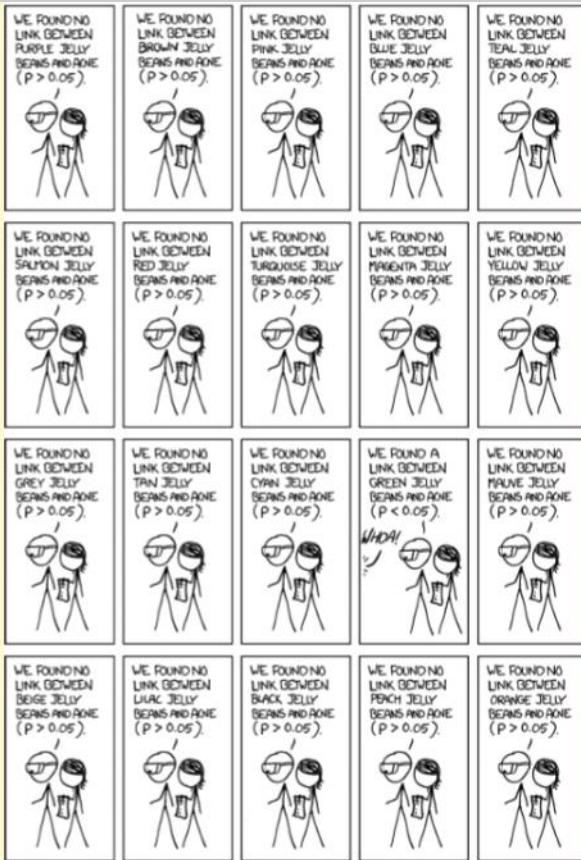
**FIGURE 9-10** The  $P$ -value for a  $z$ -test. (a) The two-sided alternative  $H_1: \mu \neq \mu_0$ . (b) The one-sided alternative  $H_1: \mu > \mu_0$ . (c) The one-sided alternative  $H_1: \mu < \mu_0$ .

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<https://xkcd.com/882>

<https://xkcd.com/882>  
<https://explainxkcd.com/882>



Questions, comments,  
or concerns?